

1. The PCB "Isolation Slot" (Creepage Air Gap) Explained

To understand why we cut a slot through the fiberglass PCB, we must look at how high-voltage electricity behaves in humid air:

BAD DESIGN (Flat Surface):

[220V AC Track] ——— (Humid Dust / Moisture Layer on PCB) ——— [5V DC Track]

▲ Micro-sparks creep across damp surface!

PRO DESIGN (Isolation Slot Cut):

[220V AC Track] ——— [AIR GAP / PHYSICAL SLOT CUT THROUGH PCB] ——— [5V DC Track]

▲ Electricity CANNOT jump across open air!

1. **Surface Creepage:** In a bathroom, high humidity creates an invisible film of microscopic water droplets and dust on the surface of your Zero PCB (perfboard). Over time, 220V AC electricity can "creep" across this damp surface toward your low-voltage 5V DC tracks. This creates a high-voltage short circuit that destroys the ESP32 or causes a fire hazard.
2. **The Air Gap Solution:** Air is a far better electrical insulator than a damp PCB surface. By taking a blade, small saw, or rotary tool (Dremel) and **cutting a 2mm-wide physical slot entirely through the fiberglass board** between the 220V AC section and the 5V DC section, you break the surface tracking path. The electricity cannot crawl across open air.

2. Complete Components List & Bangladeshi Market Alternatives

If you cannot find a specific primary component in local shops (RoboticsBD, TechshopBD, or local Patuatuli/Stadium market shops), use these drop-in alternatives:

Primary Component	Engineering Function	Local Market Alternative	Notes for 10–20 Year Build
HLK-PM01 (5V 3W)	Isolated AC-to-DC Mains Power Supply	HLK-5M05 (5V 5W) or Hi-Link HLK-10M05	Mean Well IRM-03-5 is an industrial alternative.
ESP32-C3 SuperMini	Core System Microcontroller	ESP32 DevKitC WROOM-32 or	Any ESP32 board works; DevKitC is easier for

		ESP32-C3 Zero	breadboard testing.
HLK-LD2410B Radar	24GHz FMCW Human Presence Sensor	HLK-LD2410C or HLK-LD2420	LD2410C has built-in 2.54mm headers instead of SH1.27mm.
5V Opto-Isolated Relay	Switches the 220V AC Light Load	G3MB-202P 5V Solid State Relay (SSR)	SSR is superior: SSRs switch silently with zero mechanical contacts to wear out over 20 years.
1A Glass Fuse & PCB Holder	Mains Overcurrent Protection	1A 250V Radial Micro Fuse (392 Series) or 1A PPTC Resettable Fuse	Solderable directly onto the Zero PCB.
10D471K Varistor (MOV)	Mains High-Voltage Surge Protection	7D471K or 14D471K MOV	Any 470V clamping MOV will protect against grid transients.
10Ω Inrush Limiter	Power-Up Surge Protection	10D-9 NTC Thermistor or 10Ω 2W Flameproof Metal Oxide Resistor	A 10Ω 2W flameproof resistor is widely available in local shops.
0.1μF 275VAC X2 Film Cap	RC Snubber (Relay Contact Protection)	100nF / Code 104 275V–310V AC X2 Safety Film Capacitor	Must be X2 safety-rated for AC mains across relay terminals.
100Ω 1W Resistor	RC Snubber Current Limiter	100Ω 2W Metal Film Resistor	Pair in series with the 0.1μF X2 capacitor across

			relay COM & NO.
220μF 16V Capacitor	DC Low-Voltage Noise Suppression	330μF 16V or 470μF 16V Electrolytic Capacitor	Filters 24GHz RF current spikes from the radar.
SH1.27mm Cable	Radar to PCB Wire Connection	Direct 28 AWG Stranded Copper Wires	Strip and solder directly from LD2410B pads to PCB.

3. A-to-Z LD2410B Calibration & App Guide

Step 1: App Installation & Pairing

1. Download **HLKRadarTool** from the Google Play Store or Apple App Store.
2. Enable **Bluetooth** and **Location Services (GPS)** on your smartphone (Android requires Location permissions to discover BLE devices).
3. Power up the node. Open the app, tap **Search Device**, and select **HLK-LD2410B_XXXX**.
4. Enter the default control password: **HiLink** (case-sensitive).

Step 2: Understanding Distance Gates

The LD2410B divides its detection area into **9 Distance Gates** (Gate 0 through Gate 8). Each gate represents roughly **0.75 meters (75 cm)** of depth.

[Sensor Mounted Above Door]

- └─ Gate 0 (0.0m - 0.75m) : Doorway Threshold
- └─ Gate 1 (0.75m - 1.50m): Vanity / Sink Area
- └─ Gate 2 (1.50m - 2.25m): Toilet Area <-- Maximum sensitivity needed for sitting still!
- └─ Gate 3 (2.25m - 3.00m): Shower / Far Wall
- └─ Gate 4+ : OUTSIDE BATHROOM (Set to 0 Sensitivity to prevent false triggers)

Step 3: Exact Bathroom Calibration Blueprint

1. **Set Max Detection Range:** Measure your bathroom depth. If your bathroom wall is 2.2 meters away, set **Max Moving Gate = 3** and **Max Static Gate = 3**. This mathematically blinds the sensor to people walking down the hallway outside the bathroom door.

2. Tune Gate Sensitivities:

- **Gate 0 & Gate 1 (Entrance/Sink):** Set Moving Sensitivity = **60**, Static Sensitivity = **40**.
 - **Gate 2 & Gate 3 (Toilet/Shower):** Set Moving Sensitivity = **50**, Static Sensitivity = **20** or **15**. (Note: A lower threshold number in HLKRadarTool means *HIGHER* sensitivity to micro-movement like breathing).
 - **Gates 4 through 8:** Set Moving Sensitivity = **0**, Static Sensitivity = **0**.
3. **Set Unmanned Delay:** Set **Unmanned Duration = 5 seconds**. This ensures that when you step out of the bathroom, the hardware waits 5 seconds to confirm the room is empty before turning off the relay.

4. Enterprise-Grade ESP32 Production Firmware

This code includes:

- **Hardware Watchdog Timer (esp_task_wdt):** If the MCU hangs due to an electrical glitch, the watchdog reboots the system automatically within 5 seconds.
- **Over-The-Air (OTA) Updates:** Allows you to update firmware over Wi-Fi without opening the sealed bathroom wall box.
- **Instant Hardware Sub-10ms Relay Activation:** Local radar execution happens instantly outside network callbacks.
- **Non-Blocking Asynchronous Network Layer:** Wi-Fi/MQTT failures will never freeze the light control.

```
#include <WiFi.h>
#include <PubSubClient.h>
#include <esp_now.h>
#include <ArduinoOTA.h>
#include <esp_task_wdt.h>

// --- HARDWARE PINOUT MAPPING ---
#define RADAR_OUT_PIN 8 // Connected to LD2410B OUT Pin
#define RELAY_PIN 10 // Connected to Relay Module IN Pin
#define WATCHDOG_TIMEOUT_SEC 5

// --- NETWORK CONFIGURATION ---
const char* WIFI_SSID = "YOUR_WIFI_SSID";
const char* WIFI_PASSWORD = "YOUR_WIFI_PASSWORD";
const char* MQTT_SERVER = "192.168.1.100"; // Raspberry Pi Home Assistant IP
const int MQTT_PORT = 1883;

// Master ESP32-WROOM MAC Address for ESP-NOW Secondary Backup
```

```

uint8_t masterMacAddress[] = {0x24, 0x0A, 0xC4, 0xAA, 0xBB, 0xCC};

// --- OBJECT INSTANCES ---
WiFiClient espClient;
        mqttClient        ;

// --- GLOBAL VARIABLES ---
volatile bool presenceDetected = false;
bool lastRelayState = false;
unsigned long lastPresenceTimestamp = 0;
const unsigned long OFF_TIMEOUT_MS = 3000; // 3-Second local failsafe delay
unsigned long lastNetworkRetry = 0;

typedef struct struct_message {
    char nodeID[16];
    bool lightActive;
} struct_message;
struct_message telemetryPacket;

// --- INSTANT INTERRUPT SERVICE ROUTINE (ISR) ---
void                handleRadarInterrupt {
    presenceDetected = (digitalRead(RADAR_OUT_PIN) == HIGH);
}

void transmitStatus bool {
    telemetryPacket.lightActive = state;
    // 1. ESP-NOW Direct Peer Transmission
    esp_now_send(masterMacAddress, (uint8_t *)&telemetryPacket,
sizeof(telemetryPacket));

    // 2. Local LAN MQTT Transmission
    if (mqttClient.connected()) {
        mqttClient.publish("home/bathroom/light/state", state ? "ON" : "OFF", true);
    }
}

void setup {
    Serial.begin(115200);

    // Initialize Pins
    pinMode(RELAY_PIN, OUTPUT);
    digitalWrite(RELAY_PIN, LOW); // Failsafe startup state: OFF
    pinMode(RADAR_OUT_PIN, INPUT_PULLDOWN);

```

```
// Attach Hardware Interrupt for Instant (<5ms) Light Activation
attachInterrupt(digitalPinToInterrupt(RADAR_OUT_PIN), handleRadarInterrupt,
CHANGE);
```

```
// Initialize Hardware Watchdog Timer
esp_task_wdt_init(WATCHDOG_TIMEOUT_SEC, true);
esp_task_wdt_add(NULL); // Add current thread
```

```
// Start Non-Blocking Wi-Fi
WiFi.mode(WIFI_STA);
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
```

```
// Initialize ESP-NOW
if (esp_now_init() == ESP_OK) {
    esp_now_peer_info_t peerInfo = {};
    memcpy(peerInfo.peer_addr, masterMacAddress, 6);
    peerInfo.channel = 0;
    peerInfo.encrypt = false;
    esp_now_add_peer(&peerInfo);
}
```

```
// Configure MQTT
mqttClient.setServer(MQTT_SERVER, MQTT_PORT);
strcpy(telemetryPacket.nodeID, "BATH_C3_NODE");
```

```
// Configure OTA Updates
ArduinoOTA.setHostname("Bathroom-Presence-Node");
ArduinoOTA.begin();
```

```
// Read Initial Sensor State
presenceDetected = (digitalRead(RADAR_OUT_PIN) == HIGH);
}
```

```
void loop {
    // Reset Watchdog Timer on every loop execution
    esp_task_wdt_reset();

    // --- 1. LOCAL FAILSAFE RELAY EXECUTION (Highest Priority) ---
    bool currentPresence = presenceDetected; // Read volatile ISR value

    if (currentPresence) {
        lastPresenceTimestamp = millis(); // Refresh presence timer
    }
}
```

```

    if (!lastRelayState) {
        digitalWrite(RELAY_PIN, HIGH); // ACTUATE LIGHT INSTANTLY
        lastRelayState = true;
        transmitStatus(true);
    }
} else {
    // Enforce 3-Second turn-off timeout
    if (lastRelayState && (millis() - lastPresenceTimestamp >= OFF_TIMEOUT_MS)) {
        digitalWrite(RELAY_PIN, LOW); // TURN LIGHT OFF
        lastRelayState = false;
        transmitStatus(false);
    }
}

// --- 2. OTA & BACKGROUND NETWORK MAINTENANCE ---
if (WiFi.status() == WL_CONNECTED) {
    ArduinoOTA.handle();

    if (!mqttClient.connected()) {
        if (millis() - lastNetworkRetry > 5000) { // Retry every 5s non-blocking
            lastNetworkRetry = millis();
            if (mqttClient.connect("ESP32_Bathroom_Node")) {
                transmitStatus(lastRelayState);
            }
        }
    } else {
        mqttClient.loop();
    }
}
}

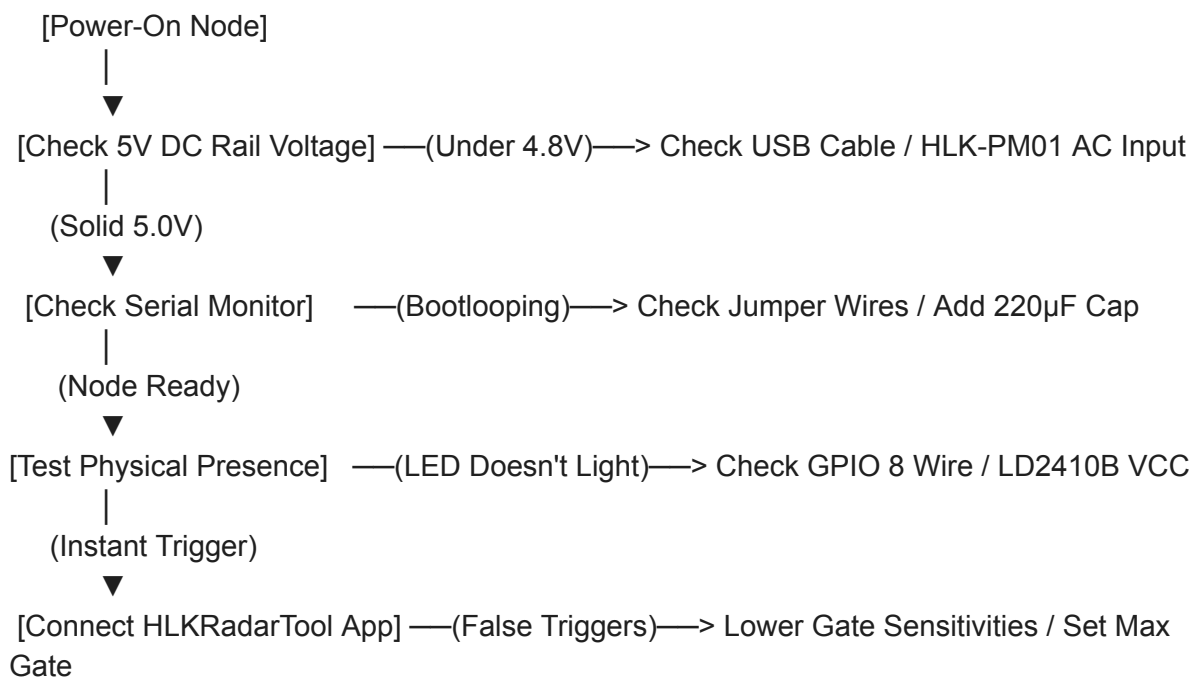
```

5. Pinout Interconnection Table

Component & Pin	ESP32-C3 Pin	Board Track Connection
HLK-LD2410B VCC	—	5V DC Power Rail (HLK-PM01 Output (+))
HLK-LD2410B GND	—	Common Ground Rail (HLK-PM01 Output (-))

HLK-LD2410B OUT	GPIO 8	Logic Input (Interrupted for Instant Detection)
Relay Module VCC	—	5V DC Power Rail
Relay Module GND	—	Common Ground Rail
Relay Module IN	GPIO 10	Logic Output (Drives Transistor/Optocoupler)
220µF Capacitor (+)	—	5V DC Power Rail
220µF Capacitor (-)	—	Common Ground Rail

6. Debugging & Fine-Tuning Procedure



1. **Power Quality Verification:** Measure voltage across the 5V and GND pins on the ESP32 using a multimeter. It must read between **4.9V and 5.2V DC**. If it drops below 4.7V when

the radar transmits, add a larger capacitor (470 μ F) across the 5V rail.

2. **False Trigger Fine-Tuning:** If the light turns on when nobody is in the bathroom:
 - Open **HLKRadarTool** app.
 - Check if a moving or static target is registering on higher gates (Gate 4, 5, or 6).
 - Reduce the **Max Gate** limit in the app until external hallway targets disappear completely.
3. **OTA Firmware Flashing Test:** Once the node is sealed inside its box, test uploading code wirelessly over ArduinoOTA via the Arduino IDE (**Tools > Port > Bathroom-Presence-Node at 192.168.X.X**). This verifies that you can tweak timeouts or code logic in the future without opening the plastic enclosure.